Συστήματα Διαχείρισης και Ανάλυσης Δεδομένων
Διδάσκων: Ιωάννης Κωτίδης

Εαρινό εξάμηνο 2024-2025

Ενδεικτικές Λύσεις Δεύτερης Εργασίας

Λύση Πρώτου Ζητήματος

```
-- Δημιουργία Σχήματος Αποθήκης Δεδομένων (Star Schema)

CREATE TABLE runners (
    runnerId INT PRIMARY KEY,
    birthYear INT,
    gender VARCHAR(1),
    country VARCHAR(3)
);

CREATE TABLE races (
    raceId INT PRIMARY KEY,
    rname VARCHAR(200),
    distance VARCHAR(10),
    country VARCHAR(3)
);

CREATE TABLE categories (
    catCode VARCHAR(4) PRIMARY KEY,
    catTitle VARCHAR(50)
);

CREATE TABLE calendar (
    raceDate DATE PRIMARY KEY,
    ryear INT,
    rmonth INT,
    rday INT
);

CREATE TABLE fact (
    raceId INT,
    raceDate DATE,
    runnerId INT,
    catCode VARCHAR(4),
    performance VARCHAR(20),
    finishTime DECIMAL(4, 2),
    averageSpeed DECIMAL(5, 3),

    PRIMARY KEY (raceId, raceDate, runnerId),
    FOREIGN KEY (raceId) REFERENCES races(raceId),
    FOREIGN KEY (runnerId) REFERENCES runners(runnerId),
    FOREIGN KEY (catCode) REFERENCES categories(catCode),
    FOREIGN KEY (raceDate) REFERENCES calendar(raceDate)
);
```

```sql
-- Εισαγωγή δεδομένων

INSERT INTO runners (runnerId, birthYear, gender, country)
SELECT DISTINCT runnerId, runnerBirthYear, runnerGender, runnerCountry
   FROM raceData;

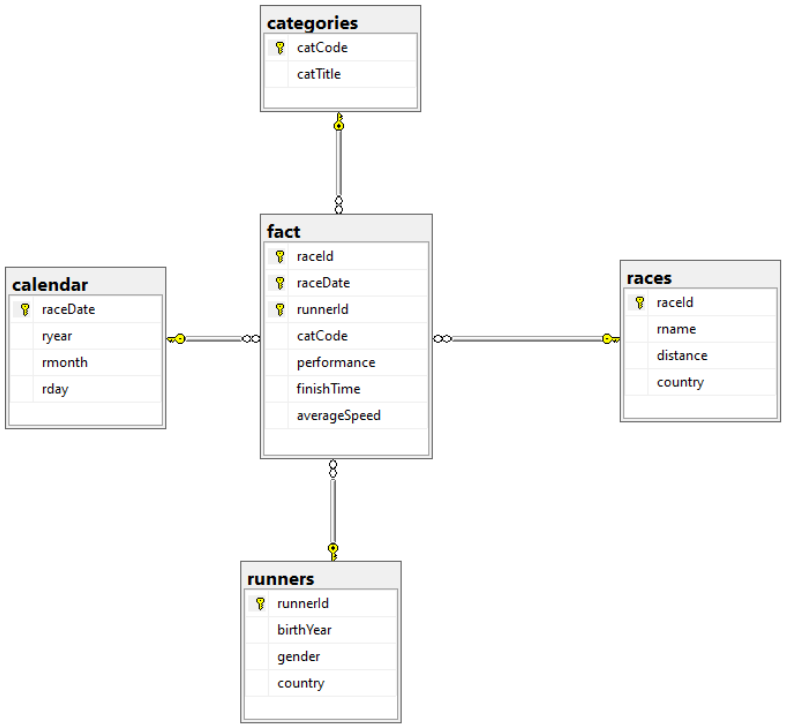
INSERT INTO races (raceId, rname, distance, country)
SELECT DISTINCT raceId, raceName, raceDistance, raceCountry
    FROM raceData
GROUP BY raceId, raceName, raceDistance, raceCountry
ORDER BY raceId;

INSERT INTO categories (catCode, catTitle)
SELECT DISTINCT ageCategoryCode, ageCategoryTitle
   FROM raceData;

INSERT INTO calendar (raceDate, ryear, rmonth, rday)
SELECT DISTINCT raceDate, YEAR(raceDate), MONTH(raceDate), DAY(raceDate)
   FROM raceData;

INSERT INTO fact (raceId, raceDate, runnerId, catCode, performance,
                  finishTime, averageSpeed)
SELECT raceId, raceDate, runnerId, ageCategoryCode, performance,
       finishTime, averageSpeed
   FROM raceData;
```

Διαγραμματική Αναπαράσταση Σχήματος Αστέρα



Λύση Δεύτερου Ζητήματος

```sql
-- Q1: Αριθμός αγώνων ανά χώρα και έτος

SELECT races.country, calendar.ryear, COUNT(DISTINCT fact.raceId)
  FROM fact JOIN races ON fact.raceId = races.raceId
            JOIN calendar ON fact.raceDate = calendar.raceDate
GROUP BY races.country, calendar.ryear
ORDER BY races.country, calendar.ryear;

-- Q2: Μέσος χρόνος τερματισμού δρομέων ανά ηλικιακή κατηγορία και φύλο σε
αγώνες 50km

SELECT categories.catTitle, runners.gender, AVG(fact.finishTime)
  FROM fact JOIN categories ON fact.catCode = categories.catCode
            JOIN runners ON fact.runnerId = runners.runnerId
            JOIN races ON fact.raceId = races.raceId
WHERE races.distance = '50km'
GROUP BY categories.catTitle, runners.gender
ORDER BY categories.catTitle, runners.gender;

-- Q3: Αριθμός Ελλήνων δρομέων που συμμετείχαν σε αγώνες

SELECT calendar.ryear, calendar.rmonth, COUNT(DISTINCT fact.runnerId)
  FROM fact JOIN calendar ON fact.raceDate = calendar.raceDate
            JOIN runners ON fact.runnerId = runners.runnerId
WHERE runners.country = 'GRE'
GROUP BY calendar.ryear DESC, calendar.rmonth DESC;
ORDER BY calendar.ryear DESC, calendar.rmonth DESC;

-- Q4: Κύβος

SELECT races.country, races.distance, runners.gender, COUNT(fact.runnerId)
  FROM fact JOIN races ON fact.raceId = races.raceId
            JOIN runners ON fact.runnerId = runners.runnerId
GROUP BY CUBE (races.country, races.distance, runners.gender);
```